A1:PROPOSAL OF QUESTION

Can we use the Market basket analysis to identify the items that are commonly purchased alongside the Apple pencils in our stores from the teleco market basket dataset?

A2: DEFINED GOAL

The aim of carrying out this analysis is to utilize the machine learning technique, market basket analysis, on the teleco market basket dataset so as to identify the items that our customers frequently purchase alongside the Apple pencils, so that we can improve the store organization for the items frequently purchased with the Apple pencils and increase sales for the telecommunication company.

B1: EXPLANATION OF MARKET BASKET

Market Basket analysis is a data mining technique that is used to discover patterns or associations between items in a large dataset. Market basket analysis identifies a combination of products that customers often purchase together, often referred to as frequent itemsets or baskets. The Market basket analysis uses the Apriori algorithm. The Apriori algorithm is a classic algorithm in data mining that is used to find frequent itemsets or groups of items that often appear together and generate association rules from large datasets. In the case of the teleco dataset, the market basket analysis will use the Apriori algorithm to

- Scan all transactions in the teleco dataset.
- Find the frequent item combinations based on a minimum support threshold. This is also called the frequent itemsets, and it is a collection of one or more items that appear together in transactions.
- Generate association rules. Association rules are the if-then statements that describe the relationship between items in a transactional dataset. Lim, Yenwee (2022, April 8) states

that association rule mining is a rule-based machine learning method that helps uncover meaningful correlations between different products according to their occurrence in a dataset.

The expected outcome from the market basket analysis is

i. We shall obtain the frequent itemsets from the teleco dataset. This includes items frequently purchased with Apple pencils.

ii. The analysis will provide strong association rules. According to Lim, Yenwee (2022, April 8), Market basket analysis is implemented using association rule mining. The association rules will reveal support, confidence, and lift rules where the apple pencils are associated with the other items.

    i. Support shows how often an itemset appears in the dataset. Support is the proportion of transactions that contain a certain itemset. support is the probability of an event occurring. It measures the proportion of transactions in which an item set appears. Support will show how frequently Apple Pencil appears in the teleco transactions.

    ii. Confidence tells you how often the consequent is true when the antecedent is true. Confidence is the likelihood that a rule is correct. Confidence is a conditional probability of a consequent event given an antecedent event.

    iii. Lift measures how much more likely the consequent is, given the antecedent, compared to chance. Lift measures how likely an item is to be purchased when another item is purchased, while controlling how popular both items are. If the Lift is greater than 1, then this is a positive association.

- Based on the findings from the basket analysis, we shall be able to make recommendations for strategies that the business can utilize to enhance the sale of the Apple pencils.

B2: TRANSACTION EXAMPLE

A transaction refers to a single customer purchase event, which is a set of items bought together at the same time. Below is an example of a transaction from the teleco dataset showing the items that were purchased in the first transaction. The transaction has 20 items that were purchased by the customer.

```
#An example of a transaction in the dataset
teleco_cleaned_df.iloc[0]
```

```
Item01                 Logitech M510 Wireless mouse
Item02                                    HP 63 Ink
Item03                                    HP 65 ink
Item04                  nonda USB C to USB Adapter
Item05                   10ft iPHone Charger Cable
Item06                                 HP 902XL ink
Item07                  Creative Pebble 2.0 Speakers
Item08           Cleaning Gel Universal Dust Cleaner
Item09                 Micro Center 32GB Memory card
Item10     YUNSONG 3pack 6ft Nylon Lightning Cable
Item11                   TopMate C5 Laptop Cooler pad
Item12                     Apple USB-C Charger cable
Item13                  HyperX Cloud Stinger Headset
Item14                   TONOR USB Gaming Microphone
Item15                 Dust-Off Compressed Gas 2 pack
Item16                 3A USB Type C Cable 3 pack 6FT
Item17                         HOVAMP iPhone charger
Item18                       SanDisk Ultra 128GB card
Item19           FEEL2NICE 5 pack 10ft Lighning cable
Item20            FEIYOLD Blue light Blocking Glasses
Name: 0, dtype: object
```

```
teleco_basket_df.iloc[0]
```

```
10ft iPHone Charger Cable                True
10ft iPHone Charger Cable 2 Pack        False
3 pack Nylon Braided Lightning Cable    False
3A USB Type C Cable 3 pack 6FT           True
5pack Nylon Braided USB C cables        False
                                        ...
iPhone 12 case                          False
iPhone Charger Cable Anker 6ft          False
iPhone SE case                          False
nonda USB C to USB Adapter               True
seenda Wireless mouse                   False
Name: 0, Length: 119, dtype: bool
```

B3: MARKET BASKET ASSUMPTION

Market Basket Analysis assumes that products purchased together in a transaction have an association or meaningful relationship, which is often interpreted as one product complementing or influencing the purchase of another.

C1: TRANSFORMING THE DATA SET

cleaned data

teleco_cleaned_df.csv

C2:CODE EXECUTION

In executing the code, the following steps were carried out

- Importing the necessary libraries

```
!pip install pandas mlxtend
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from mlxtend.frequent_patterns import apriori, association_rules
from mlxtend.preprocessing import TransactionEncoder
```

- Importing and reading the data

```
#Loading the data
#File path variables

Local_path = 'C:/Users/beatr/OneDrive/Desktop/WGU/D212/Task 3/Data/'

InputFilePath = Local_path + 'teleco_market_basket.csv'
OutputfilePath =  Local_path + 'teleco_cleaned_df.csv'


teleco_df = pd.read_csv(InputFilePath)
```

- Initial data exploration

```
teleco_df.head(100)
```

	Item01	Item02	Item03	Item04	Item05	Item06	Item07	Item08	Item09	Item10	Item11	Item12	Item13	Item14	Item15
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	Logitech M510 Wireless mouse	HP 63 Ink	HP 65 ink	nonda USB C to USB Adapter	10ft iPhone Charger Cable	HP 902XL ink	Creative Pebble 2.0 Speakers	Cleaning Gel Universal Dust Cleaner	Micro Center 32GB Memory card	YUNSONG 3pack 6ft Nylon Lightning Cable	TopMate C5 Laptop Cooler pad	Apple USB-C Charger cable	HyperX Cloud Stinger Headset	TONOR USB Gaming Microphone	Dust-Of Compressec Gas 2 pacl
2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
teleco_df.shape
```

```
(15002, 20)
```

```
teleco_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15002 entries, 0 to 15001
Data columns (total 20 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Item01  7501 non-null   object
 1   Item02  5747 non-null   object
 2   Item03  4389 non-null   object
 3   Item04  3345 non-null   object
 4   Item05  2529 non-null   object
 5   Item06  1864 non-null   object
 6   Item07  1369 non-null   object
 7   Item08  981 non-null    object
 8   Item09  654 non-null    object
 9   Item10  395 non-null    object
 10  Item11  256 non-null    object
 11  Item12  154 non-null    object
 12  Item13  87 non-null     object
 13  Item14  47 non-null     object
 14  Item15  25 non-null     object
 15  Item16  8 non-null      object
 16  Item17  4 non-null      object
 17  Item18  4 non-null      object
 18  Item19  3 non-null      object
 19  Item20  1 non-null      object
dtypes: object(20)
memory usage: 2.3+ MB
```

- Data cleaning

```
#Removing blank lines in the teleco market basket and resetting the index
teleco_cleaned_df=teleco_df.dropna(how='all').reset_index(drop=True)
```

```
teleco_cleaned_df.shape
```

```
(7501, 20)
```

```
teleco_cleaned_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7501 entries, 0 to 7500
Data columns (total 20 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Item01  7501 non-null   object
 1   Item02  5747 non-null   object
 2   Item03  4389 non-null   object
 3   Item04  3345 non-null   object
 4   Item05  2529 non-null   object
 5   Item06  1864 non-null   object
 6   Item07  1369 non-null   object
 7   Item08  981 non-null    object
 8   Item09  654 non-null    object
 9   Item10  395 non-null    object
 10  Item11  256 non-null    object
 11  Item12  154 non-null    object
 12  Item13  87 non-null     object
 13  Item14  47 non-null     object
 14  Item15  25 non-null     object
 15  Item16  8 non-null      object
 16  Item17  4 non-null      object
 17  Item18  4 non-null      object
 18  Item19  3 non-null      object
 19  Item20  1 non-null      object
dtypes: object(20)
memory usage: 1.1+ MB
```

```
teleco_cleaned_df.head()
```

	Item01	Item02	Item03	Item04	Item05	Item06	Item07	Item08	Item09	Item10	Item11	Item12	Item13	Item14	Item15
0	Logitech M510 Wireless mouse	HP 63 Ink	HP 65 ink	nonda USB C to USB Adapter	10ft iPHone Charger Cable	HP 902XL ink	Creative Pebble 2.0 Speakers	Cleaning Gel Universal Dust	Micro Center 32GB Memory	YUNSONG 3pack 6ft Nylon Lightning	TopMate C5 Laptop Cooler	Apple USB-C Charger cable	HyperX Cloud Stinger Headset	TONOR USB Gaming Microphone	Dust-Off Compressed Gas 2 pack

- Grouping the transactions

```
#Convert rows to list of non-null items
basket = teleco_cleaned_df.apply(lambda row: row.dropna().tolist(), axis=1).tolist()
#Use TransactionEncoder to get one-hot encoded dataframe
encoder = TransactionEncoder()
transactions = encoder.fit(basket).transform(basket)
# Convert to DataFrame
teleco_basket_df = pd.DataFrame(transactions, columns=encoder.columns_)
teleco_basket_df.head()
```

	10ft iPHone Charger Cable	10ft iPHone Charger Cable 2 Pack	3 pack Nylon Braided Lightning Cable	3A USB Type C Cable 3 pack 6FT	5pack Nylon Braided USB C cables	ARRIS SURFboard SB8200 Cable Modem	Anker 2-in-1 USB Card Reader	Anker 4-port USB hub	Anker USB C to HDMI Adapter	Apple Lightning to Digital AV Adapter	...	hP 65 Tri-color ink	iFixit Pro Tech Toolkit	iPhone 11 case	iPhone 12 Charger cable	iPhone 12 Pro case	iPhone 12 case	iPh Cha C Ar
0	True	False	False	True	False	False	False	False	False	False	...	False	False	False	False	False	False	F
1	False	False	False	False	False	False	False	False	False	True	...	False	False	False	False	False	False	F
2	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	F
3	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	F
4	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	F

5 rows × 119 columns

```
teleco_basket_df.iloc[0]
```

```
10ft iPHone Charger Cable                  True
10ft iPHone Charger Cable 2 Pack          False
3 pack Nylon Braided Lightning Cable      False
3A USB Type C Cable 3 pack 6FT             True
5pack Nylon Braided USB C cables          False
                                           ...
iPhone 12 case                            False
iPhone Charger Cable Anker 6ft            False
iPhone SE case                            False
nonda USB C to USB Adapter                 True
seenda Wireless mouse                     False
Name: 0, Length: 119, dtype: bool
```

- Apriori and association rules

```
# Use the Apriori algorithm to find the frequent itemsets on the teleco_market basket
#Apriori is a classic algorithm in data mining used to find frequent itemsets (groups of items that often appear together)
#and generate association rules from large datasets.Itemset is a collection of one or more items that appear together in tran
#include itemsets that appear in at least 2% of the transactions
#Return the actual item names
frequent_itemsets = apriori(teleco_basket_df, min_support = 0.02, use_colnames = True)
frequent_itemsets.sort_values(by='support',ascending=False).reset_index(drop=True)
```

	support	itemsets
0	0.238368	(Dust-Off Compressed Gas 2 pack)
1	0.179709	(Apple Pencil)
2	0.174110	(VIVO Dual LCD Monitor Desk mount)
3	0.170911	(USB 2.0 Printer cable)
4	0.163845	(HP 61 ink)
...	...	...
98	0.020397	(HP 63 Ink)
99	0.020264	(SanDisk Ultra 128GB card, Dust-Off Compressed...
100	0.020131	(Premium Nylon USB Cable, Dust-Off Compressed ...
101	0.020131	(HP 62XL Tri-Color ink, Dust-Off Compressed Ga...
102	0.020131	(USB 2.0 Printer cable, Stylus Pen for iPad)

103 rows × 2 columns

```python
#Generating association rules from the frequent itemset for items with a lift greater than 1
#Association rules are if-then statements that help discover interesting relationships between variables (items) in large dat
#Lift Measures how much more likely the consequent is, given the antecedent, compared to chance.
#Lift > 1 → positive association (they occur together more than expected)
rules = association_rules(frequent_itemsets, metric = 'lift', min_threshold = 1)
rules=rules.sort_values(by='lift', ascending=False).reset_index(drop=True)
rules.head()
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	cert
0	(VIVO Dual LCD Monitor Desk mount)	(SanDisk Ultra 64GB card)	0.174110	0.098254	0.039195	0.225115	2.291162	1.0	0.022088	1.163716	0.682343	0.168096	0.14
1	(SanDisk Ultra 64GB card)	(VIVO Dual LCD Monitor Desk mount)	0.098254	0.174110	0.039195	0.398915	2.291162	1.0	0.022088	1.373997	0.624943	0.168096	0.27
2	(FEIYOLD Blue light Blocking Glasses)	(VIVO Dual LCD Monitor Desk mount)	0.065858	0.174110	0.022930	0.348178	1.999758	1.0	0.011464	1.267048	0.535186	0.105651	0.21
3	(VIVO Dual LCD Monitor Desk mount)	(FEIYOLD Blue light Blocking Glasses)	0.174110	0.065858	0.022930	0.131700	1.999758	1.0	0.011464	1.075829	0.605334	0.105651	0.07
4	(10ft iPHone Charger Cable 2 Pack)	(Dust-Off Compressed Gas 2 pack)	0.050527	0.238368	0.023064	0.456464	1.914955	1.0	0.011020	1.401255	0.503221	0.086760	0.28

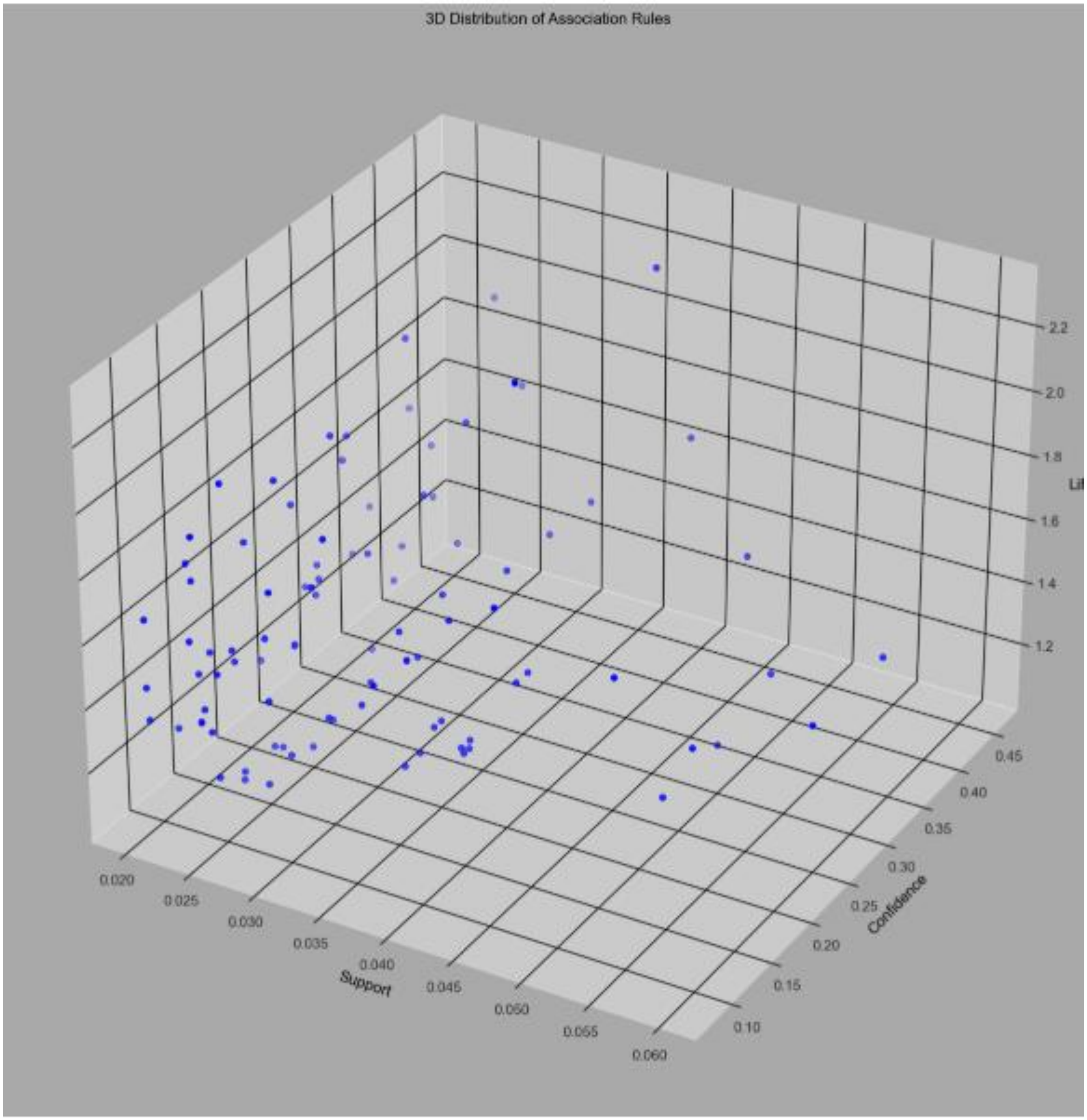
```python
print("Rules identified:", len(rules))
```

```
Rules identified: 94
```

```python
#Visualization
#Code source: Lim, Yenwee (2022, April 8) Data Mining: Market Basket Analysis with Apriori Algorithm: Visualizations
#  https://towardsdatascience.com/data-mining-market-basket-analysis-with-apriori-algorithm-970ff256a92c/

sns.set(style="whitegrid")
fig = plt.figure(figsize=(15, 15), facecolor='darkgray')
ax = fig.add_subplot(projection='3d')
x = rules['support']
y = rules['confidence']
z = rules['lift']
ax.set_facecolor('darkgray')
ax.xaxis._axinfo["grid"]['color'] = "black"
ax.yaxis._axinfo["grid"]['color'] = "black"
ax.zaxis._axinfo["grid"]['color'] = "black"
ax.set_xlabel("Support", color='black')
ax.set_ylabel("Confidence", color='black')
ax.set_zlabel("Lift", color='black')
ax.set_title("3D Distribution of Association Rules", color='black')

ax.scatter(x, y, z, color='blue')
plt.show()
```



- Finding antecedents and consequents

```
#finding the antecedent itemsets that are most common in the rules
#The antecedent is the "if" part of an association rule.
#antecedent the item or group of items that come before the arrow (→) in a rule

rules.antecedents.value_counts()
```

```
antecedents
(Dust-Off Compressed Gas 2 pack)           18
(VIVO Dual LCD Monitor Desk mount)         12
(Apple Pencil)                              9
(HP 61 ink)                                 8
(Screen Mom Screen Cleaner kit)             7
(USB 2.0 Printer cable)                     6
(Nylon Braided Lightning to USB cable)      5
(Apple Lightning to Digital AV Adapter)     4
(Apple USB-C Charger cable)                 4
(Stylus Pen for iPad)                       4
(SanDisk Ultra 64GB card)                   4
(FEIYOLD Blue light Blocking Glasses)       2
(Anker USB C to HDMI Adapter)               2
(Logitech M510 Wireless mouse)              2
(Syntech USB C to USB Adapter)              1
(TopMate C5 Laptop Cooler pad)              1
(10ft iPHone Charger Cable 2 Pack)          1
(Falcon Dust Off Compressed Gas)            1
(SanDisk Ultra 128GB card)                  1
(HP 62XL Tri-Color ink)                     1
(Premium Nylon USB Cable)                   1
Name: count, dtype: int64
```

```
#finding the consequent itemsets that are most common in the rules
#The consequent is the "then" part of an association rule.
#consequent is the item or group of items that come after the arrow (→) in a rule.
rules.consequents.value_counts()
```

```
consequents
(Dust-Off Compressed Gas 2 pack)           18
(VIVO Dual LCD Monitor Desk mount)         12
(Apple Pencil)                              9
(HP 61 ink)                                 8
(Screen Mom Screen Cleaner kit)             7
(USB 2.0 Printer cable)                     6
(Nylon Braided Lightning to USB cable)      5
(Apple Lightning to Digital AV Adapter)     4
(Apple USB-C Charger cable)                 4
(Stylus Pen for iPad)                       4
(SanDisk Ultra 64GB card)                   4
(FEIYOLD Blue light Blocking Glasses)       2
(Anker USB C to HDMI Adapter)               2
(Logitech M510 Wireless mouse)              2
(Syntech USB C to USB Adapter)              1
(TopMate C5 Laptop Cooler pad)              1
(10ft iPHone Charger Cable 2 Pack)          1
(Falcon Dust Off Compressed Gas)            1
(SanDisk Ultra 128GB card)                  1
(Premium Nylon USB Cable)                   1
(HP 62XL Tri-Color ink)                     1
Name: count, dtype: int64
```

- Getting all the rules involving the Apple Pencil

```
print("Rules identified:", len(ApplePencil_df))
```

```
Rules identified: 18
```

Apple pencil association rules sorted by confidence value in descending order

```
#getting all rules that involve Apple Pencil either as antecedent or consequent
ant_df = rules[rules['antecedents'] == {'Apple Pencil'}]
con_df = rules[rules['consequents'] == {'Apple Pencil'}]
ApplePencil_df = pd.concat([ant_df, con_df])
ApplePencil_df.sort_values(by='confidence', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	ce
0	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.
1	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.
2	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.
3	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.
4	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.
5	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.
6	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.
7	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.
8	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.
9	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.
10	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.
11	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.
12	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.
13	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.
14	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.
15	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.
16	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.
17	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.

Apple pencil association rules sorted by lift value in descending order

```
ApplePencil_df.sort_values(by='lift', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
1	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
2	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.048003	0.204510
3	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.070640	0.204510
4	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.059616	0.174766
5	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.028583	0.174766
6	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.058836	0.174446
7	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.028296	0.174446
8	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515
9	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.041372	0.248515
10	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.040521	0.207735
11	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.038128	0.207735
12	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.035300	0.206533
13	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.036684	0.206533
14	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.027910	0.193661
15	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.024963	0.193661
16	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
17	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213

Apple pencil association rules sorted by support value in descending order

```
ApplePencil_df.sort_values(by='support', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515
1	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.041372	0.248515
2	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.035300	0.206533
3	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.036684	0.206533
4	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.038128	0.207735
5	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.040521	0.207735
6	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.024963	0.193661
7	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.027910	0.193661
8	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.048003	0.204510
9	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.070640	0.204510
10	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
11	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
12	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
13	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213
14	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.028296	0.174446
15	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.028583	0.174766
16	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.059616	0.174766
17	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.058836	0.174446

Attached is the code.

To execute the code, please modify the local path variable

Local_path = 'C:/Users/beatr/OneDrive/Desktop/WGU/D212/Task 3/Data/'

BeatriceMwangi_D2
12 DataMining II_Ta

C3:ASSOCIATION RULES TABLE

Association rules table

```
#Generating association rules from the frequent itemset for items with a lift greater than 1
#Association rules are if-then statements that help discover interesting relationships between variables (items) in large datasets
#Lift Measures how much more likely the consequent is, given the antecedent, compared to chance.
#Lift > 1 → positive association (they occur together more than expected)
rules = association_rules(frequent_itemsets, metric = 'lift', min_threshold = 1)
rules=rules.sort_values(by='lift', ascending=False).reset_index(drop=True)
rules
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(VIVO Dual LCD Monitor Desk mount)	(SanDisk Ultra 64GB card)	0.174110	0.098254	0.039195	0.225115	2.291162	1.0	0.022088	1.163716	0.682343	0.168096	0.140684	0.312015
1	(SanDisk Ultra 64GB card)	(VIVO Dual LCD Monitor Desk mount)	0.098254	0.174110	0.039195	0.398915	2.291162	1.0	0.022088	1.373997	0.624943	0.168096	0.272197	0.312015
2	(FEIYOLD Blue light Blocking Glasses)	(VIVO Dual LCD Monitor Desk mount)	0.065858	0.174110	0.022930	0.348178	1.999758	1.0	0.011464	1.267048	0.535186	0.105651	0.210764	0.239939
3	(VIVO Dual LCD Monitor Desk mount)	(FEIYOLD Blue light Blocking Glasses)	0.174110	0.065858	0.022930	0.131700	1.999758	1.0	0.011464	1.075829	0.605334	0.105651	0.070484	0.239939
4	(Dust-Off Compressed Gas 2 pack)	(10ft iPhone Charger Cable 2 Pack)	0.238368	0.050527	0.023064	0.096756	1.914955	1.0	0.011020	1.051182	0.627330	0.086760	0.048690	0.276610
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
89	(Apple USB-C Charger cable)	(HP 61 ink)	0.132116	0.163845	0.023464	0.177598	1.083943	1.0	0.001817	1.016724	0.089231	0.086106	0.016449	0.160402
90	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
91	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213
92	(Screen Mom Screen Cleaner kit)	(USB 2.0 Printer cable)	0.129583	0.170911	0.023730	0.183128	1.071482	1.0	0.001583	1.014956	0.076645	0.085742	0.014735	0.160987
93	(USB 2.0 Printer cable)	(Screen Mom Screen Cleaner kit)	0.170911	0.129583	0.023730	0.138846	1.071482	1.0	0.001583	1.010756	0.080466	0.085742	0.010642	0.160987

94 rows × 14 columns

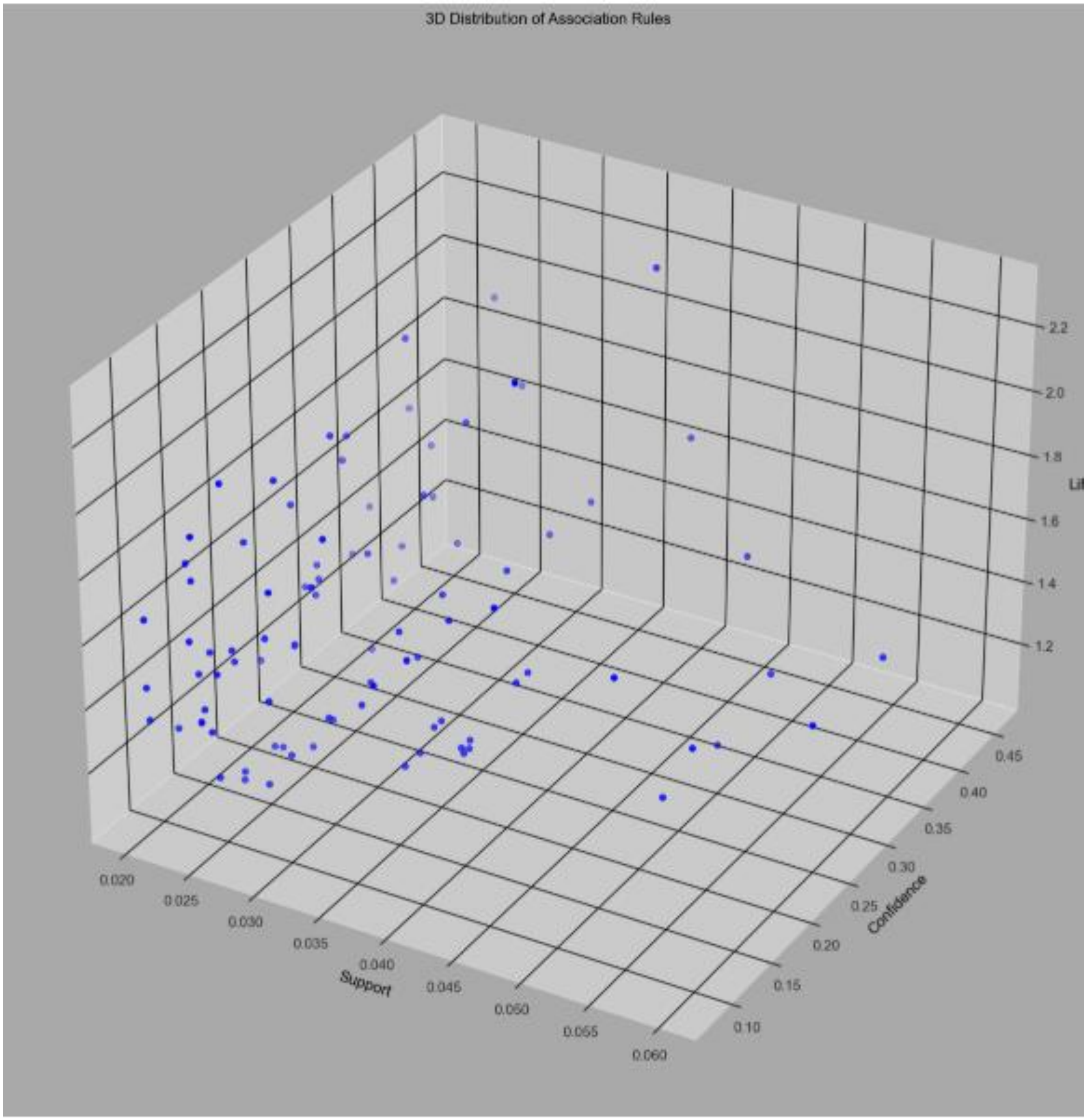
```
print("Rules identified:", len(rules))
```

```
Rules identified: 94
```

Output of the first 30 rules

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(VIVO Dual LCD Monitor Desk mount)	(SanDisk Ultra 64GB card)	0.174110	0.098254	0.039195	0.225115	2.291162	1.0	0.022088	1.163716	0.682343	0.168096	0.140684	0.312015
1	(SanDisk Ultra 64GB card)	(VIVO Dual LCD Monitor Desk mount)	0.098254	0.174110	0.039195	0.398915	2.291162	1.0	0.022088	1.373997	0.624943	0.168096	0.272197	0.312015
2	(FEIYOLD Blue light Blocking Glasses)	(VIVO Dual LCD Monitor Desk mount)	0.065858	0.174110	0.022930	0.348178	1.999758	1.0	0.011464	1.267048	0.535186	0.105651	0.210764	0.239939
3	(VIVO Dual LCD Monitor Desk mount)	(FEIYOLD Blue light Blocking Glasses)	0.174110	0.065858	0.022930	0.131700	1.999758	1.0	0.011464	1.075829	0.605334	0.105651	0.070484	0.239939
4	(Dust-Off Compressed Gas 2 pack)	(10ft iPhone Charger Cable 2 Pack)	0.238368	0.050527	0.023064	0.096756	1.914955	1.0	0.011020	1.051182	0.627330	0.086760	0.048690	0.276610
5	(10ft iPhone Charger Cable 2 Pack)	(Dust-Off Compressed Gas 2 pack)	0.050527	0.238368	0.023064	0.456464	1.914955	1.0	0.011020	1.401255	0.503221	0.086760	0.286354	0.276610
6	(Nylon Braided Lightning to USB cable)	(Screen Mom Screen Cleaner kit)	0.095321	0.129583	0.023597	0.247552	1.910382	1.0	0.011245	1.156781	0.526755	0.117219	0.135532	0.214826
7	(Screen Mom Screen Cleaner kit)	(Nylon Braided Lightning to USB cable)	0.129583	0.095321	0.023597	0.182099	1.910382	1.0	0.011245	1.106099	0.547490	0.117219	0.095921	0.214826
8	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
9	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
10	(FEIYOLD Blue light Blocking Glasses)	(Dust-Off Compressed Gas 2 pack)	0.065858	0.238368	0.027596	0.419028	1.757904	1.0	0.011898	1.310962	0.461536	0.099759	0.237201	0.267400
11	(Dust-Off Compressed Gas 2 pack)	(FEIYOLD Blue light Blocking Glasses)	0.238368	0.065858	0.027596	0.115772	1.757904	1.0	0.011898	1.056449	0.566075	0.099759	0.053433	0.267400
12	(Anker USB C to HDMI Adapter)	(VIVO Dual LCD Monitor Desk mount)	0.068391	0.174110	0.020931	0.306043	1.757755	1.0	0.009023	1.190117	0.462740	0.094465	0.159746	0.213129
13	(VIVO Dual LCD Monitor Desk mount)	(Anker USB C to HDMI Adapter)	0.174110	0.068391	0.020931	0.120214	1.757755	1.0	0.009023	1.058905	0.521973	0.094465	0.055628	0.213129
14	(Dust-Off Compressed Gas 2 pack)	(SanDisk Ultra 64GB card)	0.238368	0.098254	0.040928	0.171700	1.747522	1.0	0.017507	1.088672	0.561638	0.138413	0.081449	0.294127
15	(SanDisk Ultra 64GB card)	(Dust-Off Compressed Gas 2 pack)	0.098254	0.238368	0.040928	0.416554	1.747522	1.0	0.017507	1.305401	0.474369	0.138413	0.233952	0.294127
16	(SanDisk Ultra 64GB card)	(Screen Mom Screen Cleaner kit)	0.098254	0.129583	0.021997	0.223881	1.727704	1.0	0.009265	1.121499	0.467090	0.106865	0.108336	0.196817
17	(Screen Mom Screen Cleaner kit)	(SanDisk Ultra 64GB card)	0.129583	0.098254	0.021997	0.169753	1.727704	1.0	0.009265	1.086118	0.483903	0.106865	0.079290	0.196817
18	(VIVO Dual LCD Monitor Desk mount)	(Logitech M510 Wireless mouse)	0.174110	0.071457	0.021197	0.121746	1.703760	1.0	0.008756	1.057260	0.500143	0.094474	0.054159	0.209194
19	(Logitech M510 Wireless mouse)	(VIVO Dual LCD Monitor Desk mount)	0.071457	0.174110	0.021197	0.296642	1.703760	1.0	0.008756	1.174209	0.444850	0.094474	0.148363	0.209194
20	(Nylon Braided Lightning to USB cable)	(VIVO Dual LCD Monitor Desk mount)	0.095321	0.174110	0.027863	0.292308	1.678867	1.0	0.011267	1.167018	0.446965	0.115342	0.143115	0.226169
21	(VIVO Dual LCD Monitor Desk mount)	(Nylon Braided Lightning to USB cable)	0.174110	0.095321	0.027863	0.160031	1.678867	1.0	0.011267	1.077039	0.489605	0.115342	0.071528	0.226169
22	(Premium Nylon USB Cable)	(Dust-Off Compressed Gas 2 pack)	0.051060	0.238368	0.020131	0.394256	1.653978	1.0	0.007960	1.257349	0.416672	0.074752	0.204676	0.239354
23	(Dust-Off Compressed Gas 2 pack)	(Premium Nylon USB Cable)	0.238368	0.051060	0.020131	0.084452	1.653978	1.0	0.007960	1.036472	0.519145	0.074752	0.035189	0.239354
24	(Falcon Dust Off Compressed Gas)	(Dust-Off Compressed Gas 2 pack)	0.059992	0.238368	0.022797	0.380000	1.594172	1.0	0.008497	1.228438	0.396502	0.082729	0.185958	0.237819
25	(Dust-Off Compressed Gas 2 pack)	(Falcon Dust Off Compressed Gas)	0.238368	0.059992	0.022797	0.095638	1.594172	1.0	0.008497	1.039415	0.489364	0.082729	0.037921	0.237819
26	(Dust-Off Compressed Gas 2 pack)	(Nylon Braided Lightning to USB cable)	0.238368	0.095321	0.035729	0.149888	1.572463	1.0	0.013007	1.064189	0.477993	0.119911	0.060317	0.262357
27	(Nylon Braided Lightning to USB cable)	(Dust-Off Compressed Gas 2 pack)	0.095321	0.238368	0.035729	0.374825	1.572463	1.0	0.013007	1.218270	0.402413	0.119911	0.179164	0.262357
28	(VIVO Dual LCD Monitor Desk mount)	(Screen Mom Screen Cleaner kit)	0.174110	0.129583	0.035462	0.203675	1.571779	1.0	0.012900	1.093043	0.440468	0.132207	0.085123	0.238669
29	(Screen Mom Screen Cleaner kit)	(VIVO Dual LCD Monitor Desk mount)	0.129583	0.174110	0.035462	0.273663	1.571779	1.0	0.012900	1.137061	0.417935	0.132207	0.120540	0.238669

Visualization of the association rules



C4:TOP THREE RULES

The top three relevant association rules that were generated by the Apriori algorithm from the teleco dataset with a lift greater than 1.9 and a confidence greater than 0.3 are as below

1. Rule 1

- The Antecedent item is a SanDisk Ultra 64GB card
- The Consequent item is the VIVO Dual LCD Monitor Desk mount
- The Support is 0.0392, meaning 3.92% of all transactions contain both items
- The Confidence is 0.3989, meaning 39.89% of transactions with the antecedent item SanDisk Ultra 64GB card, also contain the consequent item VIVO Dual LCD Monitor Desk mount.
- The Lift is 2.29, meaning the consequent item, VIVO Dual LCD Monitor Desk mount, is 2.29 times more likely to be bought when the SanDisk Ultra 64GB card is bought, compared to purchasing the item VIVO Dual LCD Monitor Desk mount on its own.

The explanation of Rule 1 is that the customers who purchase the SanDisk Ultra 64GB card are strongly associated with also purchasing the VIVO Dual LCD Monitor Desk mount.

2. Rule 2

- The Antecedent item is FEIYOLD Blue light Blocking Glasses
- The Consequent item is VIVO Dual LCD Monitor Desk mount
- The Support is 0.0229, meaning 2.29% of all transactions contain both items
- The Confidence is 0.3482, meaning 34.82% of transactions with the antecedent item FEIYOLD Blue light Blocking Glasses also contain the consequent item VIVO Dual LCD Monitor Desk mount.
- Lift is 2.00, meaning there is a positive association between buying the antecedent item FEIYOLD Blue light Blocking Glasses, and the Consequent item is VIVO Dual LCD Monitor Desk mount

The explanation of rule 2 is that the customers who buy FEIYOLD Blue light Blocking Glasses often also purchase the VIVO Dual LCD Monitor Desk mount.

3. Rule 3

- The Antecedent item is 10ft iPHone Charger Cable 2 Pack
- The Consequent item is Dust-Off Compressed Gas 2 pack
- The Support is 0.0231, meaning 2.31% of transactions contain both items
- The Confidence is 0.4565, meaning 45.65% of transactions with the antecedent item 10ft iPHone Charger Cable 2 Pack also include the consequent item Dust-Off Compressed Gas 2 pack
- The Lift is 1.91, meaning there is a positive association between buying the antecedent item, 10ft iPHone Charger Cable 2 Pack, and the consequent item, Dust-Off Compressed Gas 2 pack

  The explanation of rule 3 is that the users buying the 10ft iPHone Charger Cable 2 Pack are frequently also buying Dust-Off Compressed Gas 2 pack.

```
#top three relevant rules generated by the Apriori algorithm where lift is greater thab 1.9 showing a strong association,
#with a confidence of 0.3

top_3_rules = rules[(rules['lift'] > 1.9) & (rules['confidence'] > 0.3)].sort_values(by=['lift'], ascending= False)
top_3_rules.reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(SanDisk Ultra 64GB card)	(VIVO Dual LCD Monitor Desk mount)	0.098254	0.174110	0.039195	0.398915	2.291162	1.0	0.022088	1.373997	0.624943	0.168096	0.272197	0.312015
1	(FEIYOLD Blue light Blocking Glasses)	(VIVO Dual LCD Monitor Desk mount)	0.065858	0.174110	0.022930	0.348178	1.999758	1.0	0.011464	1.267048	0.535186	0.105651	0.210764	0.239939
2	(10ft iPHone Charger Cable 2 Pack)	(Dust-Off Compressed Gas 2 pack)	0.050527	0.238368	0.023064	0.456464	1.914955	1.0	0.011020	1.401255	0.503221	0.086760	0.286354	0.276610

D1: SIGNIFICANCE OF SUPPORT, LIFT, AND CONFIDENCE SUMMARY

While analyzing the market basket analysis, the Apple pencil was identified as one of the top 3 products in the antecedents and consequent products, as illustrated below.

```
#finding the antecedent itemsets that are most common in the rules
#The antecedent is the "if" part of an association rule.
#antecedent the item or group of items that come before the arrow (→) in a rule

rules.antecedents.value_counts()
```

```
antecedents
(Dust-Off Compressed Gas 2 pack)            18
(VIVO Dual LCD Monitor Desk mount)          12
(Apple Pencil)                               9
(HP 61 ink)                                  8
(Screen Mom Screen Cleaner kit)              7
(USB 2.0 Printer cable)                      6
(Nylon Braided Lightning to USB cable)       5
(Apple Lightning to Digital AV Adapter)      4
(Apple USB-C Charger cable)                  4
(Stylus Pen for iPad)                        4
(SanDisk Ultra 64GB card)                    4
(FEIYOLD Blue light Blocking Glasses)        2
(Anker USB C to HDMI Adapter)                2
(Logitech M510 Wireless mouse)               2
(Syntech USB C to USB Adapter)               1
(TopMate C5 Laptop Cooler pad)               1
(10ft iPHone Charger Cable 2 Pack)           1
(Falcon Dust Off Compressed Gas)             1
(SanDisk Ultra 128GB card)                   1
(HP 62XL Tri-Color ink)                      1
(Premium Nylon USB Cable)                    1
Name: count, dtype: int64
```

```
consequents
(Dust-Off Compressed Gas 2 pack)            18
(VIVO Dual LCD Monitor Desk mount)          12
(Apple Pencil)                               9
(HP 61 ink)                                  8
(Screen Mom Screen Cleaner kit)              7
(USB 2.0 Printer cable)                      6
(Nylon Braided Lightning to USB cable)       5
(Apple Lightning to Digital AV Adapter)      4
(Apple USB-C Charger cable)                  4
(Stylus Pen for iPad)                        4
(SanDisk Ultra 64GB card)                    4
(FEIYOLD Blue light Blocking Glasses)        2
(Anker USB C to HDMI Adapter)                2
(Logitech M510 Wireless mouse)               2
(Syntech USB C to USB Adapter)               1
(TopMate C5 Laptop Cooler pad)               1
(10ft iPHone Charger Cable 2 Pack)           1
(Falcon Dust Off Compressed Gas)             1
(SanDisk Ultra 128GB card)                   1
(Premium Nylon USB Cable)                    1
(HP 62XL Tri-Color ink)                      1
Name: count, dtype: int64
```

Further filtering out the rules that had the Apple pencil as either antecedent or a consequent, there were 18 association rules as below, ordered by the confidence

```python
#getting all rules that involve Apple Pencil either as antecedent or consequent
ant_df = rules[rules['antecedents'] == {'Apple Pencil'}]
con_df = rules[rules['consequents'] == {'Apple Pencil'}]
ApplePencil_df = pd.concat([ant_df, con_df])
ApplePencil_df.sort_values(by='confidence', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
1	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515
2	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.070640	0.204510
3	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.059616	0.174766
4	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.058836	0.174446
5	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.041372	0.248515
6	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.040521	0.207735
7	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.036684	0.206533
8	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.035300	0.206533
9	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.027910	0.193661
10	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.038128	0.207735
11	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213
12	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.024963	0.193661
13	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.048003	0.204510
14	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
15	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
16	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.028296	0.174446
17	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.028583	0.174766

```python
print("Rules identified:", len(ApplePencil_df))
```

```
Rules identified: 18
```

The market basket analysis reveals that confidence values in the apple pencil association rules range from 33.02 % to 12.09 %.

Looking at the top two confidence metrics, Rule 1 has the Apple Lightning to Digital AV Adapter as the antecedent and Apple Pencil as the consequent, at 33.02% and Rule 2 has the Apple Pencil as the antecedent and Dust-Off Compressed Gas 2 pack as the consequent, rule 2 has confidence of 28.33%.

Based on the high confidence levels, there is a strong chance of the products in rule 1 and rule 2 being purchased together.

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
1	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515

Ordering the Apple Pencil association rules by the lift metric, the lift values are greater than one and range from 1.837 to 1.072. This means that there is a positive association between the Apple Pencil and the various products.

Rule 1 has the Apple Lightning to Digital AV Adapter as the antecedent and Apple Pencil as the consequent, with a lift of 1.8378. Rule 2 has the Apple Pencil as the antecedent and the Apple Lightning to Digital AV Adapter as the consequent with a lift of 1.8378.

Since the lift value for all the rules is greater than 1, it indicates that the two items are purchased together 83.78% of the time.

```
ApplePencil_df.sort_values(by='lift', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
1	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
2	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.048003	0.204510
3	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.070640	0.204510
4	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.059616	0.174766
5	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.028583	0.174766
6	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.058836	0.174446
7	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.028296	0.174446
8	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515
9	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.041372	0.248515
10	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.040521	0.207735
11	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.038128	0.207735
12	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.035300	0.206533
13	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.036684	0.206533
14	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.027910	0.193661
15	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.024963	0.193661
16	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
17	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213

Ordering the Apple Pencil association rules by the support metric, the support values range from 0.0509 to 0.0217, showing how frequently these item pairs appear together in the dataset.

Rule 1 has the Apple Pencil as the antecedent and Dust-Off Compressed Gas 2 pack as the consequent, with a support of 0.0509 (or 5.09%). Rule 2 has Dust-Off Compressed Gas 2 pack as the antecedent and Apple Pencil as the consequent, also yielding a support of 0.0509.

The support values for the two rules indicate that the combination appears in approximately 5.09% of all transactions.

```
ApplePencil_df.sort_values(by='support', ascending=False).reset_index(drop=True)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(Apple Pencil)	(Dust-Off Compressed Gas 2 pack)	0.179709	0.238368	0.050927	0.283383	1.188845	1.0	0.008090	1.062815	0.193648	0.138707	0.059103	0.248515
1	(Dust-Off Compressed Gas 2 pack)	(Apple Pencil)	0.238368	0.179709	0.050927	0.213647	1.188845	1.0	0.008090	1.043158	0.208562	0.138707	0.041372	0.248515
2	(Apple Pencil)	(VIVO Dual LCD Monitor Desk mount)	0.179709	0.174110	0.036528	0.203264	1.167446	1.0	0.005239	1.036592	0.174852	0.115126	0.035300	0.206533
3	(VIVO Dual LCD Monitor Desk mount)	(Apple Pencil)	0.174110	0.179709	0.036528	0.209801	1.167446	1.0	0.005239	1.038081	0.173666	0.115126	0.036684	0.206533
4	(Apple Pencil)	(USB 2.0 Printer cable)	0.179709	0.170911	0.036395	0.202522	1.184961	1.0	0.005681	1.039640	0.190286	0.115825	0.038128	0.207735
5	(USB 2.0 Printer cable)	(Apple Pencil)	0.170911	0.179709	0.036395	0.212949	1.184961	1.0	0.005681	1.042232	0.188267	0.115825	0.040521	0.207735
6	(Apple Pencil)	(HP 61 ink)	0.179709	0.163845	0.033196	0.184718	1.127397	1.0	0.003751	1.025603	0.137757	0.106959	0.024963	0.193661
7	(HP 61 ink)	(Apple Pencil)	0.163845	0.179709	0.033196	0.202604	1.127397	1.0	0.003751	1.028711	0.135143	0.106959	0.027910	0.193661
8	(Apple Pencil)	(Screen Mom Screen Cleaner kit)	0.179709	0.129583	0.030796	0.171365	1.322437	1.0	0.007509	1.050423	0.297236	0.110579	0.048003	0.204510
9	(Screen Mom Screen Cleaner kit)	(Apple Pencil)	0.129583	0.179709	0.030796	0.237654	1.322437	1.0	0.007509	1.076009	0.280119	0.110579	0.070640	0.204510
10	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
11	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
12	(Apple Pencil)	(Apple USB-C Charger cable)	0.179709	0.132116	0.025463	0.141691	1.072479	1.0	0.001721	1.011156	0.082387	0.088920	0.011033	0.167213
13	(Apple USB-C Charger cable)	(Apple Pencil)	0.132116	0.179709	0.025463	0.192735	1.072479	1.0	0.001721	1.016135	0.077869	0.088920	0.015879	0.167213
14	(Apple Pencil)	(Nylon Braided Lightning to USB cable)	0.179709	0.095321	0.021730	0.120920	1.268559	1.0	0.004600	1.029121	0.258084	0.085789	0.028296	0.174446
15	(Apple Pencil)	(Stylus Pen for iPad)	0.179709	0.095054	0.021730	0.120920	1.272118	1.0	0.004648	1.029424	0.260773	0.085880	0.028583	0.174766
16	(Stylus Pen for iPad)	(Apple Pencil)	0.095054	0.179709	0.021730	0.228612	1.272118	1.0	0.004648	1.063395	0.236378	0.085880	0.059616	0.174766
17	(Nylon Braided Lightning to USB cable)	(Apple Pencil)	0.095321	0.179709	0.021730	0.227972	1.268559	1.0	0.004600	1.062514	0.234010	0.085789	0.058836	0.174446

D2: PRACTICAL SIGNIFICANCE OF FINDINGS

The confidence values in the Apple Pencil association rules range from 33.02% to 12.09%, indicating varying degrees of predictive strength between product pairs. In Rule 1where the antecedent is Apple Lightning to Digital AV Adapter and the consequent item, Apple Pencil, has the highest confidence at 33.02%, meaning that approximately 33% of customers who buy the Apple Lightning to Digital AV Adapter also buy the Apple Pencil. The second rule, where the Apple Pencil is the antecedent and the Dust-Off Compressed Gas 2 pack is the consequent, has a confidence of 28.33%, showing that nearly 28% of Apple Pencil buyers also purchase the Dust-Off Compressed Gas 2 pack.

The relatively high confidence scores imply strong customer buying behavior correlations, suggesting that cross-selling or bundling these product pairs could significantly increase overall sales.

The lift values for the Apple Pencil association rules range from 1.8378 down to 1.072, where all lift values were exceeding 1. This signals a positive association between the products beyond random chance.

Rule 1 has the Apple Lightning to Digital AV Adapter as the antecedent and Apple Pencil as the consequent, with a lift of 1.8378, indicating these two items are purchased together nearly 83.78% more often than would be expected if they were independent.

Rule 2 has the Apple Pencil as the antecedent and the Apple Lightning to Digital AV Adapter as the consequent, also yielding a lift of 1.8378.

The strong lift reinforces the potential for strategic placement of these products together in physical or online stores to leverage natural purchasing patterns and increase basket size.

Support values measure how frequently the product pairs appear together in total transactions, with the top Apple Pencil association rules showing support around 5.09% and as low as 2.17%.

Rule 1 has the Apple Pencil as the antecedent and Dust-Off Compressed Gas 2 pack as the consequent, holds a support of 0.0509, meaning this combination occurs in just over 5% of all transactions. Rule 2 has Dust-Off Compressed Gas 2 pack as the antecedent and Apple Pencil as the consequent, also yielding a support of 0.0509.

The support values highlight that these item pairs are statistically significant and also practically relevant due to their regular occurrence in customer purchases. The support rate justifies prioritizing these pairs in inventory planning and marketing efforts to meet existing customer demand effectively.

D3: COURSE OF ACTION

From the results of the market basket analysis for the product Apple Pencil in the teleco basket dataset,18 association rules were generated. The results for the confidence, lift, and support metrics are summarized below,

1. The Confidence values ranged from 33.02% to 12.09%, with the Apple Lightning to Digital AV Adapter showing the highest confidence (33.02%) when predicting Apple Pencil purchases, and the Dust-Off Compressed Gas 2 pack following closely at 28.33%. This indicates a strong likelihood that these items are bought together.
2. The lift values, ranging from 1.837 to 1.072, further confirm these associations occur more often than would be expected by chance alone, particularly with a lift of 1.8378 for both top rules involving the Apple Lightning to Digital AV Adapter and Apple Pencil.
3. The support values ranged from 5.09% to 2.17%, showing that these combinations are also relatively frequent in overall transactions, with the highest being the Apple Pencil and Dust-Off Compressed Gas 2 pack at 5.09%.

Based on the findings, I recommend that the telecommunication company strategically implement targeted cross-promotional strategies that bundle the Apple Pencil with the Apple Lightning to Digital AV Adapter and the Dust-Off Compressed Gas 2 pack, since these items emerged as the strongest associations among the 18 rules involving the Apple Pencil. The telecommunication company should prioritize placing the products together on the aisle in the stores, make recommendations to the online sales customers for items frequently purchased together, and tailor marketing promotional bundles for these products so as to increase sales.

E: PANOPTO VIDEO OF CODE

https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=2c44da9c-483c-45b6-968b-b30f018171a1

E1: PANOPTO VIDEO OF PROGRAMS

https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=2c44da9c-483c-45b6-968b-b30f018171a1

F: SOURCES FOR THIRD-PARTY CODE

Lim, Yenwee (2022, April 8) Data Mining: Market Basket Analysis with Apriori Algorithm: Visualizations

https://towardsdatascience.com/data-mining-market-basket-analysis-with-apriori-algorithm-970ff256a92c/

G: SOURCES

Lim, Yenwee (2022, April 8) Data Mining: Market Basket Analysis with Apriori Algorithm

https://towardsdatascience.com/data-mining-market-basket-analysis-with-apriori-algorithm-970ff256a92c/

H : PROFESSIONAL COMMUNICATION